



## Jargon Buster

**Artefacts and noise:** Marks and smudges picked up in the course of scanning or data transfer that do not correspond to the original document. Artefacts might include pixellation, dotted lines, regularly repeated patterns, moiré, etc.

**Batch scanning:** Some scanners can automatically recognise several images placed on it and save them as separate files. Also, an automatic document feeder can be used to send pages one at a time to be scanned, saving a lot of time.

**Bit depth:** The scanner stores a certain number of bits of data for each pixel in an image: 1 bit represents two colours (black and white). Similarly, 2 bits represent four colours—black, two shades of grey and white. The higher the number of bits devoted for this (bit depth), the better the scanner is at distinguishing between really close shades of the same colour. This results in better image quality.

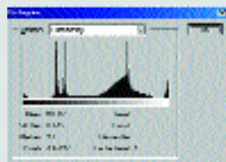
**Contrast:** Contrast determines the gap between the darkest shade and lightest shade (in effect the number of shades actually used in an image). An image with low contrast tends to look lifeless and flat.

**CCD array:** Known as Charge Coupled Device array, this is a set of light-sensitive diodes used in scanners and digital

cameras. It sweeps across a document during capture and, when exposed to light, generates a series of digital signals that are converted into pixel values.

**Halftone:** Most printing processes are not capable of printing continuous tone images. Hence, dots of varying size or density are used to give the impression of grey areas in a greyscale image.

**Histogram:** An imaging tool that shows how pixels are distributed by tonal value in



*The histogram function in Photoshop*

an image. Each grey level in a greyscale image has a vertical line that shows the number of pixels. Usually, a histogram shows low values (dark parts of the images) on the left, climbing to the highest and lightest on the right. A histogram is a useful method for checking that you have a good scan, with a full range of colours.

**Hue and saturation:** The aspect of colour that distinguishes it from another colour (what makes a colour red or green or blue). Hue is distinct from saturation, which measures the intensity of the hue (more

red, more green, etc).

**Midtones:** An image is logically divided into highlights (bright areas), midtones (medium-bright areas) and shadows (dark areas). If converted to greyscale, midtones correspond to 50 per cent grey.

**Resolution:** The resolution, measured in dots or pixels per inch (dpi or ppi), defines the ability of an imaging device to capture or display individual dots or picture elements (pixels). There are two types of resolution: optical and interpolated, of which the first is more important. More pixels mean better resolution and sharper image quality. The resolution can be increased to capture finer details, but after a point, added resolution will not result in an appreciable gain in image quality but only results in a larger file size.

**Tonal range:** Also called the dynamic range or the optical density (OD) of a scanner, it is a measure of how well the scanner can capture the gradations from bright highlights to dark shadows in an image.

**TWAIN:** It is a software industry standard that allows TWAIN-compliant applications and scanners to communicate. It allows the scanner to be accessed from within another program, for example an image editing or DTP application.

adapters for both 35 mm and wide format films, used largely by professionals. HP ScanJet 5400C only had the interface for attaching the adapter, whereas its elder sibling the HP ScanJet 5470C had the adapter too.

Power saving is another critical feature—if the CCF (cold cathode fluorescent) tube is not in use for a specified period, the scanner shuts off automatically. This not only saves power but also enhances the life of the CCF.

In the Budget range, only three scanners—BenQ SW2 4300U, SW2 5000E and Umax AstraSlim 600—had this feature.



*Epson Perfection 1650 Photo offers the maximum number of features from one-touch controls to excellent bundled software and even a transparency adapter*

Among the mid range scanners, only Agfa SnapScan 1212P and Mustek 1200ED did not have provision for power saving, while all the scanners in the high-end category possessed power saving features.

A transport lock protects the scanner from damage during transporting. In the budget range all three BenQ scanners, the two Canon scanners and the Microtek ScanMaker 3800 had the transport lock feature. In the mid range, the transport lock feature was available only in two

models—BenQ ScanPrisa 640BT and Microtek Scanmaker X6EL. Among high-end scanners, the transport lock was missing only in the Umax Astra 4000U.

A removable lid becomes an important consideration if you need to scan images from thick magazines or books, as its absence would adversely affect the quality of the scanned images. Among the budget scanners, only the Microtek ScanMaker 3800 and 3730 sported this feature. In the mid range segment again only the Microtek series and the Umax Astra 5400 supported the removable lid feature. In the high-end segment, all five scanners had support for removable lids—not surprising as these scanners are used by professionals who need to scan from a variety of source documents.

**Overall:** In the budget category, Microtek Scanmaker 3800 came out to be the clear winner with an aggregate of 18 points, edging out its nearest rival, the BenQ 4300U. Canon CanoScan D646U had the lowest score for features here, managing only 10.75 points, mainly because its driver did not support a majority of the functions for image enhancement, nor did it have one-touch control buttons for easier scanning.

In the mid range segment, Microtek's ScanMaker X6EL scraped in as a winner with just one point more than its nearest rivals, BenQ ScanPrisa 640BT and Umax Astra 5400. It scored higher mainly because of its support for legal paper size—a feature which will be appreciated by lawyers and tax consultants. The Mustek 1200ED could only manage